

ou 1m



SPH4U V2

Chapter 7 Electricity Quiz (TI) V2

Name: _____

For each question, draw a diagram, show your formula and all units, and keep answers to 3 SDs.

1. What is the net electric force on charge d? Include a FBD. (4 marks)

$$F_{E_{db}} = \frac{k q_d q_b}{r^2}$$

$$= \frac{(8.99 \times 10^9 \text{ N m}^2/\text{C}^2)(3.0 \times 10^{-6} \text{ C})(6.2 \times 10^{-6} \text{ C})}{(0.075 \text{ m})^2}$$

$$F_{E_{db}} = 29.7 \text{ N [E]}$$

$$F_{E_{da}} = \frac{(8.99 \times 10^9 \text{ N m}^2/\text{C}^2)(3.0 \times 10^{-6} \text{ C})(4.0 \times 10^{-6} \text{ C})}{(0.07 \text{ m})^2}$$

$$F_{E_{da}} = 22.01 \text{ N [W 40° S]}$$

F_{E_{dy}}:

$$\sin 40 = \frac{y}{F_{E_{da}}}$$

$$F_{E_{dy}} = 14.14 \text{ N [S]}$$

F_{E_{dx}}:

$$\cos 40 = \frac{x}{F_{E_{da}}}$$

$$F_{E_{dx}} = 16.9 \text{ N [W]}$$

$$F_{E_{dc}} = \frac{(8.99 \times 10^9 \text{ N m}^2/\text{C}^2)(3.0 \times 10^{-6} \text{ C})(4.6 \times 10^{-6} \text{ C})}{(0.07 \text{ m})^2}$$

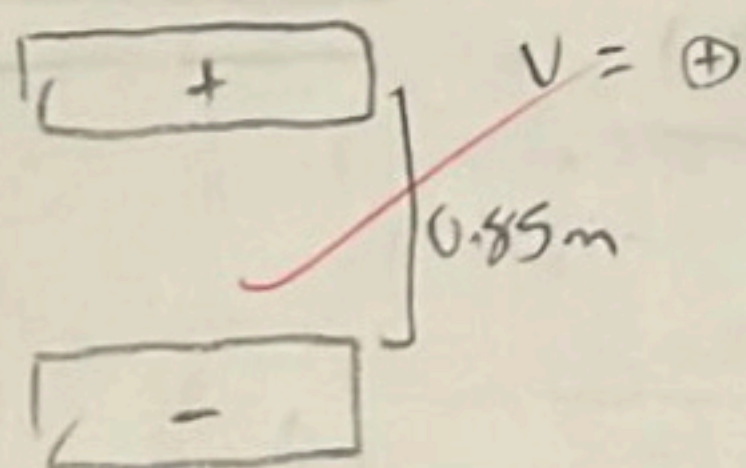
$$F_{E_{dc}} = 22.01 \text{ N [E 40° S]}$$

$$F_{E_{dcy}}: \sin 40 = \frac{y}{F_{E_{dc}}} = 14.14 \text{ N [S]}$$

$$F_{E_{dcx}}: \cos 40 = \frac{x}{F_{E_{dc}}} = 16.9 \text{ N [E]}$$

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2. Two charged plates have a potential difference of 200V and are separated by 0.85 m. What is the electric field between the plates? (2 marks)



$$V = 200 \text{ V}$$

$$d = 0.85 \text{ m}$$

$$\Delta V = -E \Delta d$$

$$\frac{V}{d} = -E$$

$$\frac{200 \text{ V}}{0.85 \text{ m}} = -E$$

$$235.3 = \frac{-E}{-1}$$

$$E = -235 \text{ N/C}$$



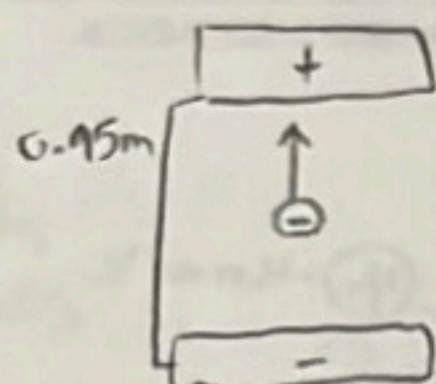
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3. A positively charged plate is 0.95 m away from a negatively charged plate. An electron starts at rest moves through a potential difference of + 890 V. What is its final velocity? (3 marks)



$$\begin{aligned} V &= \oplus \quad \Delta V = 890 \text{ V} \\ W &= \oplus \quad d = 0.95 \text{ m} \\ \Delta EE &= \ominus \quad q = -1.6 \times 10^{-19} \text{ C} \end{aligned}$$

$$E_{Ki} + E_{Pi} = E_{Kf} + E_{Pf}$$

$$E_{Ki} = \frac{1}{2}mv^2$$

$$\sqrt{\frac{2E_{Ki}}{m}} = v_f$$

$$\Delta V = \frac{\Delta EE}{q} \rightarrow \Delta EE = \Delta V q$$

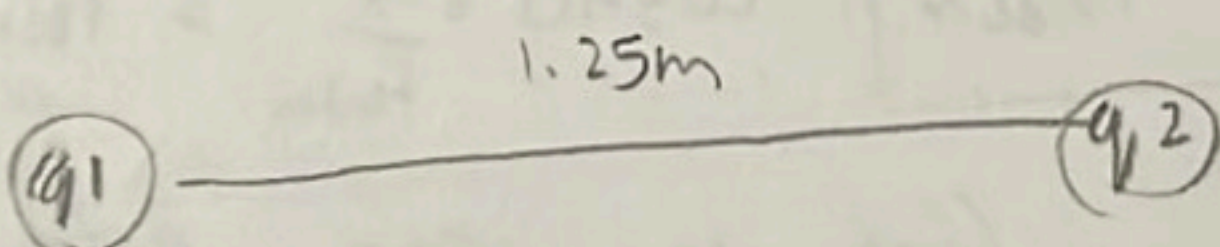
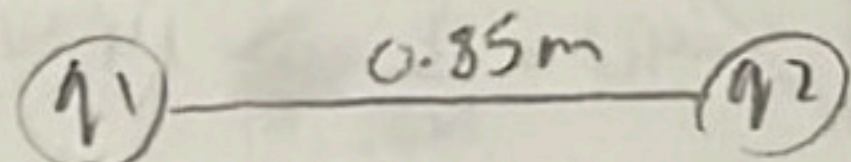
$$\Delta EE = (890 \text{ V})(-1.6 \times 10^{-19} \text{ C})$$

$$\Delta EE = -1.4124 \times 10^{-16} \text{ N}$$

$$v_f = \sqrt{\frac{2(-1.4124 \times 10^{-16} \text{ N})}{9.11 \times 10^{-31} \text{ kg}}}$$

$$v_f = 1.77 \times 10^7 \text{ m/s}$$

4. Charge 1 ($q_1 = 3.4 \times 10^{-6} \text{ C}$) is fixed at rest 0.85 m west of Charge 2 ($q_2 = 5.2 \times 10^{-6} \text{ C}$) which is also at rest. Charge 2 moves to 1.25 m away from charge 1, what is its final translational (linear) velocity if Charge 2 has a mass of $7.25 \times 10^{-15} \text{ kg}$? (3 marks)



$$v_f^2 = v_i^2 + 2a\Delta d$$

$$v_f = \sqrt{2a\Delta d}$$

$$= \sqrt{2(3.17 \times 10^{13} \text{ m/s}^2)(0.4 \text{ m})}$$

$$v_f = 5.04 \times 10^6 \text{ m/s}$$

$$F_E = \frac{kq_1q_2}{r^2}$$

$$= \frac{(8.99 \times 10^9 \text{ N m}^2/\text{C}^2)(3.4 \times 10^{-6} \text{ C})(5.2 \times 10^{-6} \text{ C})}{(0.85 \text{ m})^2}$$

$$F_E = 0.23 \text{ N}$$

$$F_E = \frac{ma}{m}$$

$$a = \frac{0.23 \text{ N}}{7.25 \times 10^{-15} \text{ kg}}$$

$$a = 3.17 \times 10^{13} \text{ m/s}^2$$

$$a = 3.17 \times 10^{13} \text{ m/s}^2$$



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Q1 cont:

$$F_{\text{net}} = F_{\text{ex}} + F_{\text{ey}}$$

$$F_{\text{ex}} = F_{\text{Edb}} + F_{\text{Edc}} + F_{\text{Edc}}$$

$$= 29.7 \text{ N} + (-16.7 \text{ N}) + 16.8 \text{ N}$$

$$F_{\text{ex}} = 29.7 \text{ N [E]}$$

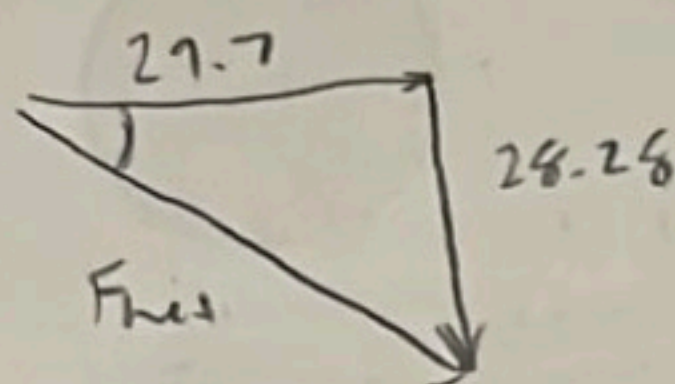
$$F_{\text{ey}} = F_{\text{Edb}} + F_{\text{Edc}} + F_{\text{Edc}}$$

$$= -14.14 \text{ N} + (-14.14 \text{ N})$$

$$F_{\text{ey}} = 28.28 \text{ N [S]}$$

$$F_{\text{net}} = \sqrt{(29.7 \text{ N})^2 + (28.28 \text{ N})^2}$$

$$F_{\text{net}} = 41.0 \text{ N [E} 44^{\circ} \text{ S]}$$



$$\tan \theta = \frac{28.28}{29.7}$$

$$\theta \approx 44^{\circ}$$

$$V_1 = 2.99 \times 10^8 \text{ m/s}$$

$$V_2 = 1.69 \times 10^8 \text{ m/s}$$



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